WASP-135b

R-0013

Benchmark run · workflow W8-benchmark · record `2026-bench_WASP-135_260630` · created 2026-07-08T03:40:03+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461221.856 +/- 0.016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.051
Transit depth (Rp/Rs)^2	2.99 +/- 1.76 [%]
Orbital Inclination (inc)	86.37 +/- 2.93
Airmass coefficient 1 (a1)	1.49 +/- 0.22
Airmass coefficient 2 (a2)	-0.326 +/- 0.069
Scatter in the residuals of the lightcurve fit is	14.67 %
Best Comparison Star	None
Optimal Aperture	5.37
Optimal Annulus	7.16
Transit Duration (day)	0.078 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.051 vs 0.139 (deviation 0.62σ)
Duration fit vs published	0.078 d vs 0.0687 d (deviation 0.61σ)
Mid-time O–C	-41.6 min
Depth SNR	3.1
Residual scatter	14.67 %

Light curve

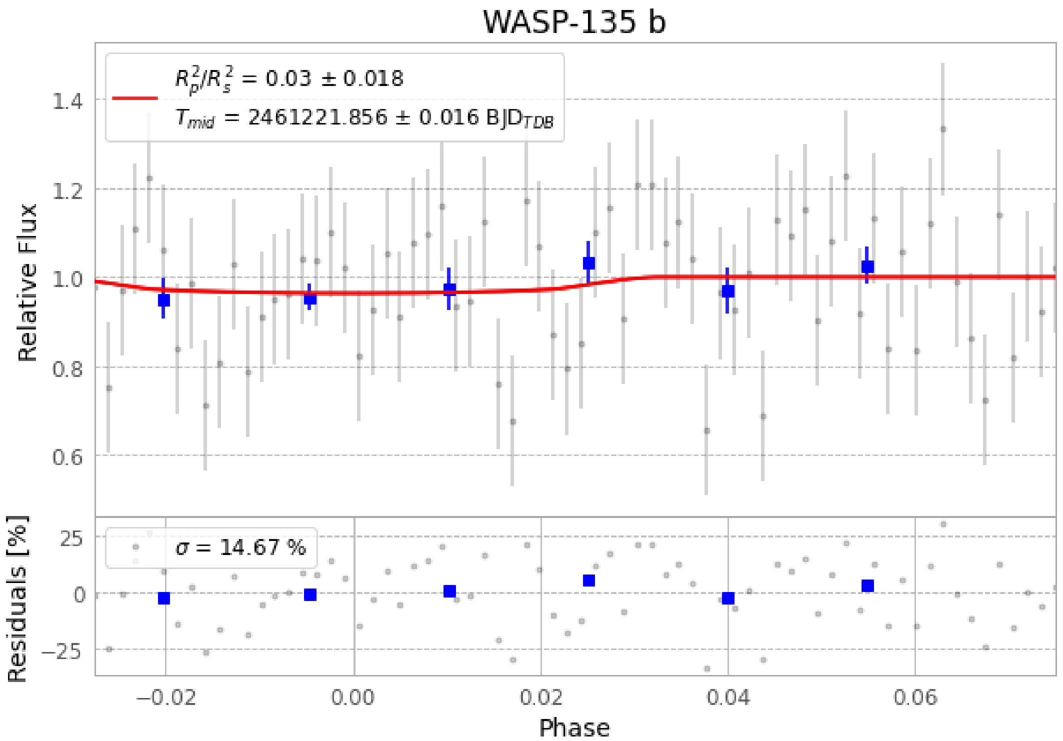

Figure 1 — FinalLightCurve_WASP-135 b_2026-06-30.png (SHA-256 5b30e9b3c11588c2…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 174], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2ee37d647ec2fd20…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

70 FITS frames (46 MB), `WASP-135260630073028.FITS` → `WASP-135260630105715.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-260630.log (db3438d78ba8…), FinalLightCurve_WASP-135 b_2026-06-30.png (5b30e9b3c115…), FinalLightCurve_WASP-135 b_2026-06-30.csv (44426bd4249f…)

Compute resources

Wall time 131.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)